

Elementary Problems Functional Equation Solutions

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Basic Work

1. Replacing x by $y - 2$, we obtain

$$f(y) = 2(y - 2)^2 - 3(y - 2) + 1 = 2y^2 - 11y + 15.$$

As y can be any real number, this is the function we need. No checking is needed since the step is reversible.

2. Replacing x by $\frac{y+1}{3}$, we obtain

$$f(y) = 4\left(\frac{y+1}{3}\right) + 8 = \frac{4y+28}{3}.$$

As y can be any real number, this is the function we need. No checking is needed since the step is reversible.

3. For any $y \geq 1$, we can take $x = \sqrt{y-1}$. This gives

$$f(y) = (\sqrt{y-1})^2 + 2 = y + 1.$$

In other words, the functions that we are looking for must satisfy $f(y) = y + 1$ for $y \geq 1$, while there is no restriction on $f(y)$ for $y < 1$. For any such function, one easily checks that the equation $f(x^2 + 1) = x^2 + 2$ is satisfied for any $x \in \mathbb{R}$. Thus, there are infinitely many solutions.

4. After expansion, the equation becomes $2f(x) + x + 2 = 0$. This can be rewritten as

$$f(x) = -\frac{x+2}{2}.$$

This is the only function that satisfies the equation as the steps are reversible.

5. Putting $x = \frac{3}{2}$ in the equation, we obtain the contradiction $0 = -1$. Therefore, no such function exists.
6. The equation can be rewritten as

$$(4x^2 - 1)f(x) = 0.$$

For $x \neq \pm\frac{1}{2}$, this yields $f(x) = 0$. Conversely, for any function satisfying $f(x) = 0$ for $x \neq \pm\frac{1}{2}$, the given equation is satisfied for any $x \in \mathbb{R}$, including $x = \pm\frac{1}{2}$. Therefore, any such function is a solution. In other words, there is no restriction on $f\left(\pm\frac{1}{2}\right)$.

7. The equation can be rewritten as $(f(x) - 4)(f(x) + 1) = 0$. Thus, for each $x \in \mathbb{R}$, we have $f(x) = 4$ or $f(x) = -1$. Since each equation involves only one $f(x)$, the images of different x 's are independent. This means we may have $f(x) = 4$ for some x and $f(y) = -1$ for some y . Any such function is a solution by simple checking. Thus, there are infinitely many solutions.

8. We label the given equation

$$f(x) + 3f(1 - x) = -4x + 10. \quad (1)$$

Replacing x by $1 - x$ in (1), we obtain

$$f(1 - x) + 3f(x) = -4(1 - x) + 10 = 4x + 6. \quad (2)$$

From (1) and (2), we find that

$$f(x) = \frac{3(3f(x) + f(1 - x)) - (f(x) + 3f(1 - x))}{8} = \frac{3(4x + 6) - (-4x + 10)}{8} = 2x + 1.$$

Checking:

$$f(x) + 3f(1 - x) = (2x + 1) + 3(2(1 - x) + 1) = -4x + 10.$$

9. We label the given equation

$$f(x) + 2xf\left(\frac{1}{x}\right) = x + 2. \quad (1)$$

Replacing x by $\frac{1}{x}$ in (1), we obtain

$$f\left(\frac{1}{x}\right) + \frac{2}{x}f(x) = \frac{1}{x} + 2. \quad (2)$$

Multiplying (2) by $2x$ and then subtracting (1), we find that

$$3f(x) = 2x\left(\frac{1}{x} + 2\right) - (x + 2) = 3x.$$

Hence, $f(x) = x$ for any $x \in \mathbb{R}^+$.

Checking:

$$f(x) + 2xf\left(\frac{1}{x}\right) = x + 2x \cdot \frac{1}{x} = x + 2.$$

10. We label the given equation

$$2f(x - 1) + 3f(1 - x) = 20. \quad (1)$$

Replacing x by $2 - x$ in (1), we obtain

$$2f(1 - x) + 3f(x - 1) = 20. \quad (2)$$

From (1) and (2), we find that

$$f(x - 1) = \frac{3(2f(1 - x) + 3f(x - 1)) - 2(2f(x - 1) + 3f(1 - x))}{5} = 4.$$

Hence, $f(x - 1) = 4$ for any $x \in \mathbb{R}$. This means $f(x)$ is the constant function 4.

Checking:

$$2f(x - 1) + 3f(1 - x) = 2(4) + 3(4) = 20.$$

11. We label the given equation

$$f(x) + f\left(\frac{x - 1}{x}\right) = \frac{x^2 + x - 1}{x}. \quad (1)$$

For any $y \neq 1$, we replace x by $\frac{1}{1 - y}$ in (1). Since

$$\frac{x - 1}{x} = (1 - y)\left(\frac{1}{1 - y} - 1\right) = y,$$

we obtain

$$f\left(\frac{1}{1 - y}\right) + f(y) = (1 - y)\left(\frac{1}{(1 - y)^2} + \frac{1}{1 - y} - 1\right) = \frac{-y^2 + y + 1}{1 - y}.$$

Changing the variable, this means

$$f\left(\frac{1}{1 - x}\right) + f(x) = \frac{-x^2 + x + 1}{1 - x} \quad (2)$$

for any $x \neq 1$.

Similarly, for any $y \neq 0, 1$, we replace x by $\frac{1}{1 - y}$ in (2). Since

$$\frac{1}{1 - x} = \left(1 - \frac{1}{1 - y}\right)^{-1} = \frac{y - 1}{y},$$

we obtain

$$f\left(\frac{y - 1}{y}\right) + f\left(\frac{1}{1 - y}\right) = \frac{y - 1}{y}\left(-\frac{1}{(1 - y)^2} + \frac{1}{1 - y} + 1\right) = \frac{-y^2 + 3y - 1}{y(1 - y)}.$$

Changing the variable, this means

$$f\left(\frac{x - 1}{x}\right) + f\left(\frac{1}{1 - x}\right) = \frac{-x^2 + 3x - 1}{x(1 - x)} \quad (3)$$

for any $x \neq 0, 1$.

Note that (1), (2) and (3) are three equations in three unknowns $f(x)$, $f\left(\frac{x-1}{x}\right)$ and $f\left(\frac{1}{1-x}\right)$. Thus, we solve the equations to get

$$f(x) = \frac{(1) + (2) - (3)}{2} = x$$

for any $x \neq 0, 1$.

For $f(0)$ and $f(1)$, the only case for which these are involved in the equation (1) is $x = 1$, where we have $f(1) + f(0) = 1$. This means $f(0)$ and $f(1)$ can be any value with sum 1.

Checking: for any $x \neq 0, 1$, we must have $\frac{x-1}{x} \neq 0, 1$. Thus,

$$f(x) + f\left(\frac{x-1}{x}\right) = x + \frac{x-1}{x} = \frac{x^2 + x - 1}{x}.$$

12. We label the given equation

$$f(xy) = xf(x) + yf(y). \quad (1)$$

Putting $x = y = 0$ in (1), we obtain $f(0) = 0$. Putting $y = 0$ in (1), we obtain $0 = xf(x)$. For nonzero x , this implies $f(x) = 0$. Together with $f(0) = 0$, the only possibility is $f(x) = 0$ for any $x \in \mathbb{R}$.

Checking:

$$f(xy) = 0 = xf(x) + yf(y).$$

13. **Solution 1:**

We label the given equation

$$f(f(x+y)) = f(x) + y. \quad (1)$$

Putting $x = 0$ in (1), we obtain $f(f(y)) = f(0) + y$. Using this to eliminate the iterated function in (1), we have

$$f(0) + (x+y) = f(f(x+y)) = f(x) + y.$$

This yields $f(x) = x + c$ for some constant $c = f(0)$.

Checking:

$$f(f(x+y)) = f(x+y+c) = x+y+2c$$

and

$$f(x) + y = x + c + y.$$

They are equal for any x, y if and only if $c = 0$. Thus, the only solution is $f(x) = x$ for any $x \in \mathbb{R}$.

Solution 2:

We label the given equation as above. Swapping x and y , we have

$$f(f(y+x)) = f(y) + x.$$

Comparing this with (1), we find that $f(x)+y = f(y)+x$. This shows $f(x)-x = f(y)-y$ for any $x, y \in \mathbb{R}$ so that $f(x) - x$ is a constant. The remaining steps are the same as solution 1.

14. We label the given equation

$$f(x-y) = f(x) + f(y) - 2xy. \quad (1)$$

Putting $x = y = 0$ in (1), we obtain $f(0) = 0$. Putting $x = y$ in (1), we obtain $2f(x) = 2x^2$. Thus, $f(x) = x^2$ for any $x \in \mathbb{R}$.

Checking:

$$f(x-y) = (x-y)^2 = x^2 + y^2 - 2xy = f(x) + f(y) - 2xy.$$

15. We label the given equation

$$f(x^2 - y^2) = x^2 - f(y^2). \quad (1)$$

Putting $x = y = 0$ in (1), we obtain $f(0) = 0$. Putting $x = y$ in (1), we obtain

$$f(x^2) = x^2.$$

This shows $f(x) = x$ for any $x \geq 0$. Now, (1) becomes

$$f(x^2 - y^2) = x^2 - y^2.$$

Note that $x^2 - y^2$ runs through all real numbers. Thus, $f(x) = x$ for any $x \in \mathbb{R}$.

Checking:

$$f(x^2 - y^2) = x^2 - y^2 = x^2 - f(y^2).$$

16. We label the given equation

$$f(x)f(y) - f(x+y) = x+y. \quad (1)$$

Putting $y = 0$ in (1), we obtain

$$f(x)f(0) - f(x) = x. \quad (2)$$

Since this holds for any x , we cannot have $f(0) = 1$. Putting $x = 0$ in (2), we obtain $f(0)^2 = f(0)$. As $f(0) \neq 1$, this yields $f(0) = 0$. Thus, (2) implies $f(x) = -x$ for any $x \in \mathbb{R}$.

Checking:

$$f(x)f(y) - f(x+y) = (-x)(-y) + (x+y) = xy + x + y.$$

This is not equal to $x + y$. Therefore, there is no solution.

17. We label the given equation

$$xf(y) + yf(x) = (x+y)f(x)f(y). \quad (1)$$

Putting $x = y$ in (1), we have $2xf(x) = 2xf(x)^2$. For $x \neq 0$, this implies $f(x) = f(x)^2$. Thus, $f(x) = 0$ or $f(x) = 1$.

If there exists $f(a) = 0$ for some $a \neq 0$, then we put $x = a$ in (1). This gives $af(y) = 0$. As $a \neq 0$, we have $f(y) = 0$ for any $y \in \mathbb{R}$. This is clearly a solution.

If $f(x) \neq 0$ for all $x \neq 0$, then $f(x) = 1$ for all $x \neq 0$ from above. We check that all such functions satisfy (1) (which means $f(0)$ can take any value). Indeed, for $x, y \neq 0$, we have

$$xf(y) + yf(x) = x + y = (x+y)f(x)f(y).$$

For $x \neq 0$ and $y = 0$, we have

$$xf(y) + yf(x) = xf(0) = (x+y)f(x)f(y).$$

Lastly, for $x = y = 0$, we have

$$xf(y) + yf(x) = 0 = (x+y)f(x)f(y).$$

Thus, (1) holds for any $x, y \in \mathbb{R}$.

For the first case, $f(y) = 0$ for all $y \in \mathbb{R}$ is continuous. For the second case, f is continuous if and only if $f(0) = 1$. Thus, there are exactly two such continuous functions.

18. **Solution 1:**

We label the given equation

$$f(x+y) + xy = f(x)f(y). \quad (1)$$

Putting $y = 0$ in (1), we obtain

$$f(x) = f(x)f(0).$$

If $f(0) \neq 1$, then $f(x) = 0$ for all $x \in \mathbb{R}$. However, this is clearly not a solution. Hence, $f(0) = 1$.

Putting $x = 1$ and $y = -1$ in (1), we obtain $f(1)f(-1) = 0$. Hence, $f(1) = 0$ or $f(-1) = 0$.

If $f(1) = 0$, by putting $y = 1$ in (1), we have $f(x+1) + x = 0$. This is the same as $f(x) = -x + 1$.

If $f(-1) = 0$, by putting $y = -1$ in (1), we have $f(x-1) - x = 0$. This is the same as $f(x) = x + 1$.

Checking: for $f(x) = \pm x + 1$, we have

$$f(x+y) + xy = \pm(x+y) + 1 + xy = (\pm x + 1)(\pm y + 1) = f(x)f(y).$$

Solution 2:

We label the given equation as above. Putting $y = 1$ in (1), we obtain

$$f(x+1) = f(1)f(x) - x. \quad (2)$$

Replacing x by $x+1$ in (2), we have

$$f(x+2) = f(1)f(x+1) - x - 1 = f(1)^2 f(x) - f(1)x - x - 1. \quad (3)$$

Putting $y = 2$ in (1), we obtain

$$f(x+2) = f(2)f(x) - 2x. \quad (4)$$

Comparing (3) and (4), we have

$$f(1)^2 f(x) - f(1)x - 1 = f(2)f(x) - x$$

for any $x \in \mathbb{R}$.

If $f(1)^2 = f(2)$, then $-f(1)x - 1 = -x$ for any $x \in \mathbb{R}$. This is impossible.

If $f(1)^2 \neq f(2)$, then $f(x)$ is a linear polynomial. Let $f(x) = ax + b$. Then (1) becomes

$$a(x+y) + b + xy = (ax+b)(ay+b).$$

This holds for any $x, y \in \mathbb{R}$ if and only if $a^2 = 1$, $ab = a$ and $b^2 = b$. Thus, $b = 1$ and $a = \pm 1$. This gives the two solutions $f(x) = \pm x + 1$ for any $x \in \mathbb{R}$.

19. We label the given equation

$$xf(y) - yf(x) = f\left(\frac{y}{x}\right). \quad (1)$$

Putting $y = 0$ in (1), we obtain $(x-1)f(0) = 0$. Since this holds for any x , this yields $f(0) = 0$. Putting $x = y$ in (1), we obtain $f(1) = 0$. Putting $y = 1$ in (1), we obtain

$$-f(x) = f\left(\frac{1}{x}\right).$$

Replacing x by $\frac{1}{x}$ in (1), we have

$$\frac{1}{x}f(y) - yf\left(\frac{1}{x}\right) = f(xy),$$

i.e.

$$\frac{1}{x}f(y) + yf(x) = f(xy). \quad (2)$$

Swapping x and y in (2), we obtain

$$\frac{1}{y}f(x) + xf(y) = f(yx). \quad (3)$$

Comparing (2) and (3), we have

$$\frac{1}{x}f(y) + yf(x) = \frac{1}{y}f(x) + xf(y) \quad (4)$$

for any $x, y \neq 0$. Putting $x = 2$ and $y = -1$ in (4), we have $f(-1) = 0$. For $x, y \neq \pm 1$, as $x - \frac{1}{x} \neq 0$, we can rewrite (4) as

$$\frac{x}{x^2 - 1}f(x) = \frac{y}{y^2 - 1}f(y).$$

This shows $\frac{x}{x^2 - 1}f(x) = c$ for some constant c , i.e. $f(x) = \frac{c(x^2 - 1)}{x}$. The same formula holds for $x = \pm 1$. So the solutions are $f(0) = 0$ and $f(x) = \frac{c(x^2 - 1)}{x}$ for $x \neq 0$.

Checking: for $y = 0$, we have

$$xf(y) - yf(x) = 0 - 0 = 0 = f\left(\frac{y}{x}\right).$$

For $y \neq 0$, we have

$$xf(y) - yf(x) = \frac{cx(y^2 - 1)}{y} - \frac{cy(x^2 - 1)}{x} = \frac{c(y^2 - x^2)}{xy} = \frac{c\left(\frac{y^2}{x^2} - 1\right)}{\frac{y}{x}} = f\left(\frac{y}{x}\right).$$

20. We label the given equation

$$f(xf(x) + f(y)) = x^2 + y. \quad (1)$$

Putting $x = 0$ in (1), we obtain

$$f(f(y)) = y. \quad (2)$$

Replacing x by $f(x)$ in (1), we obtain

$$f(f(x)f(f(x)) + f(y)) = f(x)^2 + y.$$

This yields

$$f(f(x)x + f(y)) = f(x)^2 + y.$$

Comparing this with (1), we have $f(x)^2 = x^2$. Thus, $f(x) = \pm x$ for each x .

Suppose there exist nonzero $a, b \in \mathbb{R}$ such that $f(a) = a$ and $f(b) = -b$. Putting $x = a$ and $y = b$ in (1), we obtain $f(a^2 - b) = a^2 + b$. This yields $\pm(a^2 - b) = a^2 + b$. This contradicts $a \neq 0$ and $b \neq 0$. Thus, either $f(x) = x$ for all $x \in \mathbb{R}$ or $f(x) = -x$ for all $x \in \mathbb{R}$.

Checking: for $f(x) = \pm x$, we have

$$f(xf(x) + f(y)) = f(\pm x^2 \pm y) = x^2 + y.$$

21. We label the given equation

$$f(f(y + f(x))) = f(x + y) + f(x) + y. \quad (1)$$

Replacing y by $f(y)$ in (1), we obtain

$$f(f(f(y) + f(x))) = f(x + f(y)) + f(x) + f(y). \quad (2)$$

Swapping x and y , we get

$$f(f(f(x) + f(y))) = f(y + f(x)) + f(y) + f(x). \quad (3)$$

Comparing (2) and (3), we find that

$$f(x + f(y)) = f(y + f(x)). \quad (4)$$

Next, swapping x and y in (1), we have

$$f(f(x + f(y))) = f(y + x) + f(y) + x.$$

Note that the left-hand side is the same as the left-hand side of (1) from (4). Thus, we have

$$f(x) + y = f(y) + x.$$

This shows $f(x) - x$ is a constant. Let $f(x) - x = c$. Then (1) becomes

$$x + y + 3c = (x + y + c) + (x + c) + y.$$

This cannot hold for all $x, y \in \mathbb{R}$. Thus, there is no solution.

22. We label the given equation

$$f(y) + f(x + f(y)) = y + f(f(x) + f(f(y))). \quad (1)$$

Replacing x by $f(x)$ in (1), we obtain

$$f(y) + f(f(x) + f(y)) = y + f(f(f(x)) + f(f(y))). \quad (2)$$

Swapping x and y , we get

$$f(x) + f(f(y) + f(x)) = x + f(f(f(y)) + f(f(x))). \quad (3)$$

Comparing (2) and (3), we find that

$$f(y) - y = f(x) - x.$$

Thus, $f(x) - x = c$ for some constant c . Then (1) becomes

$$(y + c) + (x + y + 2c) = y + (x + y + 4c).$$

This holds for all $x, y \in \mathbb{R}$ if and only if $c = 0$. This means the only solution is the identity function.

23. We label the given equation

$$f(x + f(y + z)) + f(f(x + y) + z) = 2y. \quad (1)$$

Putting $x = z = 0$ in (1), we obtain

$$f(f(y)) = y.$$

Putting $x = 0$ and replacing y by $f(y)$ in (1), we get

$$f(f(f(y) + z)) + f(f(f(y)) + z) = 2f(y).$$

Using $f(f(x)) = x$, this can be simplified to $f(y) + z + f(y + z) = 2f(y)$, i.e.

$$f(y + z) = f(y) - z. \quad (2)$$

Swapping y and z in (2), we obtain

$$f(z + y) = f(z) - y. \quad (3)$$

Comparing (2) and (3), we have

$$f(y) + y = f(z) + z.$$

This shows $f(y) + y = c$ for some constant c , i.e. $f(y) = c - y$ for any $y \in \mathbb{R}$.

Checking:

$$\begin{aligned} f(x + f(y + z)) + f(f(x + y) + z) &= f(x + c - y - z) + f(c - x - y + z) \\ &= (y + z - x) + (x + y - z) = 2y. \end{aligned}$$

24. We label the given equation

$$f(x + f(y)) = 2f(xf(y)). \quad (1)$$

We shall choose x and y such that $x + f(y) = xf(y)$. Note that f cannot be the constant 1 function. Thus, there exists $y \in \mathbb{R}$ such that $f(y) \neq 1$. Then we put $x = \frac{f(y)}{f(y) - 1}$ in (1) so that $x + f(y) = xf(y)$. This yields

$$f(xf(y)) = 0$$

with $x = \frac{f(y)}{f(y) - 1}$. Thus, $f(a) = 0$ for some $a \in \mathbb{R}$.

Putting $y = a$ in (1), we obtain $f(x) = 2f(0)$. Also, by putting $x = 0$, we get $f(0) = 0$. This shows $f(x) = 0$ for any $x \in \mathbb{R}$. Clearly, this is a solution.

25. We label the given equation

$$f(2f(f(x)) + y) = 2f(x + 3) + f(3y - 1). \quad (1)$$

We shall choose x and y such that $2f(f(x)) + y = 3y - 1$. Thus, we put $y = \frac{2f(f(x)) + 1}{2}$ in (1). This yields $2f(x + 3) = 0$. As this holds for any $x \in \mathbb{R}$, $f(x) = 0$ for any $x \in \mathbb{R}$. Clearly, this is a solution.

26. We label the given equation

$$f(f(x + y) - x)f(f(x + y) - y) = xy. \quad (1)$$

Suppose $f(a) = 0$ for some $a \in \mathbb{R}$. Putting $x = -a$ and $y = 2a$ in (1), we obtain

$$f(f(a) + a)f(f(a) - 2a) = -2a^2$$

and hence $-2a^2 = 0$. This yields $a = 0$.

Putting $y = 0$ in (1), we obtain

$$f(f(x) - x)f(f(x)) = 0.$$

Thus, for each $x \in \mathbb{R}$, we have $f(f(x) - x) = 0$ or $f(f(x)) = 0$. From above, this implies $f(x) - x = 0$ or $f(x) = 0$. The latter case implies $x = 0$, which can be included in the former case. Hence, $f(x) = x$ for any $x \in \mathbb{R}$.

Checking: for $f(x) = x$, we have

$$f(f(x + y) - x)f(f(x + y) - y) = (x + y - x)(x + y - y) = xy.$$

Cauchy Equation

27. We label the given equation

$$f(x + y) = f(x) + f(y). \quad (1)$$

Putting $x = y = 0$ in (1), we get $f(0) = 2f(0)$, i.e. $f(0) = 0$.

Consider any $f(t) = c$. We claim that $f(nt) = cn$ for any $n \in \mathbb{Z}$. Indeed, suppose $f(kt) = ck$ for some $k \in \mathbb{Z}$. By putting $x = kt$ and $y = t$ in (1), we have

$$f((k+1)t) = f(kt) + f(t) = ck + c = c(k+1).$$

Similarly, by putting $x = (k-1)t$ and $y = t$ in (1), we have

$$f((k-1)t) = f(kt) - f(t) = ck - c = c(k-1).$$

By induction (forward for positive integers and backward for negative integers), we have

$$f(nt) = cn = nf(t) \quad (2)$$

for any $n \in \mathbb{Z}$.

Let $f(1) = d$. Then $f(n) = dn$ for $n \in \mathbb{Z}$ from above. For any $x = \frac{m}{n} \in \mathbb{Q}$ where $n \neq 0$, we put $t = \frac{m}{n}$ in (2). This implies $f(m) = nf\left(\frac{m}{n}\right)$ and hence

$$f\left(\frac{m}{n}\right) = \frac{f(m)}{n} = \frac{dm}{n} = d \cdot \frac{m}{n}.$$

This shows $f(x) = dx$ for all $x \in \mathbb{Q}$.

Checking:

$$f(x + y) = d(x + y) = dx + dy = f(x) + f(y).$$

28. We label the given equation

$$f(x + y) = f(x) + f(y). \quad (1)$$

By Cauchy equation, we have $f(x) = cx$ for all $x \in \mathbb{Q}$ where c is a constant. Note that $c \geq 0$ since $f(x) \geq 0$ for any $x \geq 0$. Suppose $y \geq x$. Replacing y by $y - x$ in (1), we obtain

$$f(y) = f(x) + f(y - x) \geq f(x)$$

since $y - x \geq 0$. This shows f is an increasing function.

Consider any $x \notin \mathbb{Q}$. For any $\varepsilon > 0$, there exists rational numbers a, b satisfying $x - \varepsilon < a < x < b < x + \varepsilon$. As f is increasing, we have

$$c(x - \varepsilon) \leq ca = f(a) \leq f(x) \leq f(b) = cb \leq c(x + \varepsilon).$$

Taking sufficiently small $\varepsilon > 0$, this forces $f(x) = cx$. Thus, this holds for any $x \in \mathbb{R}$.

Checking:

$$f(x + y) = c(x + y) = cx + cy = f(x) + f(y).$$

29. We label the given equation

$$f(xy) = f(x)f(y). \quad (1)$$

Putting $x = y$ in (1), we obtain $f(x^2) = f(x)^2 \geq 0$. This shows $f(x) \geq 0$ for any $x \geq 0$. By Cauchy equation, the only possibility is $f(x) = cx$ for any $x \in \mathbb{R}$ where $c \geq 0$ is a constant. In that case, (1) becomes $cxy = (cx)(cy)$. This holds for any $x, y \in \mathbb{R}$ if and only if $c = c^2$, i.e. $c = 0, 1$. Therefore, f is the zero function or the identity function.

Checking: for $f(x) = cx$ where $c = 0, 1$, we have

$$f(x + y) = c(x + y) = cx + cy = f(x) + f(y)$$

and

$$f(xy) = cxy = c^2xy = (cx)(cy) = f(x)f(y).$$

30. Let $g(x) = f(x) - 3$ for any $x \in \mathbb{Q}$. Then the equation becomes

$$g(x + y) + 3 + 3 = g(x) + 3 + g(y) + 3,$$

i.e. $g(x + y) = g(x) + g(y)$. By Cauchy equation, the solutions are $g(x) = cx$ for some $c \in \mathbb{Q}$. Therefore, $f(x) = cx + 3$ for any $x \in \mathbb{Q}$.

Checking:

$$f(x + y) + 3 = c(x + y) + 3 + 3 = (cx + 3) + (cy + 3) = f(x) + f(y).$$

31. Let $g(x) = f(x) + x^2$ for any $x \in \mathbb{Q}$. Then the equation becomes

$$g(x + y) - (x + y)^2 + 2xy = g(x) - x^2 + g(y) - y^2,$$

i.e. $g(x + y) = g(x) + g(y)$. By Cauchy equation, the solutions are $g(x) = cx$ for some $c \in \mathbb{Q}$. Therefore, $f(x) = cx - x^2$ for any $x \in \mathbb{Q}$.

Checking:

$$f(x + y) + 2xy = c(x + y) - (x + y)^2 + 2xy = (cx - x^2) + (cy - y^2) = f(x) + f(y).$$

32. We label the given equation

$$f(x) + f(y) = 2f\left(\frac{x + y}{2}\right). \quad (1)$$

Putting $y = 0$ in (1), we obtain

$$f(x) + f(0) = 2f\left(\frac{x}{2}\right).$$

Replacing x by $x + y$, we get

$$f(x + y) + f(0) = 2f\left(\frac{x + y}{2}\right).$$

Comparing this with (1), we have

$$f(x) + f(y) = f(x + y) + f(0). \quad (2)$$

Let $g(x) = f(x) - c$ for any $x \in \mathbb{Q}$ where $c = f(0)$ is a constant. Then (2) becomes

$$g(x) + c + g(y) + c = g(x + y) + c + c,$$

i.e. $g(x + y) = g(x) + g(y)$. By Cauchy equation, the solutions are $g(x) = ax$ for some $a \in \mathbb{Q}$. Therefore, $f(x) = ax + c$ for any $x \in \mathbb{Q}$.

Checking:

$$f(x) + f(y) = (ax + c) + (ay + c) = 2\left(a\left(\frac{x + y}{2}\right) + c\right) = 2f\left(\frac{x + y}{2}\right).$$

33. We label the given equation

$$f(2f(x) + f(y)) = 2x + y. \quad (1)$$

Putting $x = y = 0$ in (1), we obtain $f(3f(0)) = 0$. Putting $x = y = 3f(0)$ in (1), we obtain $f(0) = 9f(0)$. Thus, $f(0) = 0$.

Putting $x = 0$ in (1), we have $f(f(y)) = y$. Replacing x by $f(x)$ and y by $f(y)$ in (1), we find that

$$f(2f(f(x)) + f(f(y))) = 2f(x) + f(y).$$

Thus,

$$f(2x + y) = 2f(x) + f(y). \quad (2)$$

Putting $y = 0$ in (2), we have $f(2x) = 2f(x)$. Hence, (2) can be rewritten as

$$f(2x + y) = f(2x) + f(y).$$

We can replace x by $\frac{x}{2}$ to obtain $f(x + y) = f(x) + f(y)$ for any $x, y \in \mathbb{Q}$. By Cauchy equation, the solutions are $f(x) = cx$ for some constant $c \in \mathbb{Q}$. In that case, (1) becomes

$$c(2cx + cy) = 2x + y.$$

This holds for any $x, y \in \mathbb{Q}$ if and only if $c = \pm 1$. Thus, there are two solutions, namely, $f(x) = \pm x$ for any $x \in \mathbb{Q}$.

Integers and Rationals

34. We label the given equation

$$f(m + nf(m)) = m + f(m)f(n). \quad (1)$$

Putting $m = n = 0$ in (1), we obtain $f(0) = f(0)^2$. Thus, $f(0) = 0$ or 1 .

If $f(0) = 0$, by putting $n = 0$ in (1), we get $f(m) = m$ for any $m \in \mathbb{Z}$.

If $f(0) = 1$, by putting $n = 0$ in (1), we get $0 = m$ for any $m \in \mathbb{Z}$. This is impossible.

Thus, the only possibility is the identity function.

Checking:

$$f(m + nf(m)) = m + nm = m + f(m)f(n).$$

35. We label the given equation

$$f(m + n)^2 = f(m^2) + f(n^2). \quad (1)$$

Putting $n = -m$ in (1), we obtain $2f(m^2) = f(0)^2$. Thus, $f(m^2) = c$ for some constant $c \in \mathbb{Z}$. Putting $m = n = 0$ in (1), we obtain $c^2 = 2c$. Thus, $c = 0, 2$.

Using $f(m^2) = c$, we can rewrite (1) as $f(m + n)^2 = 2c$. As $m + n$ runs through all integers, this implies $f(n) = \pm\sqrt{2c}$ for any $n \in \mathbb{Z}$. In other words, if $c = 0$, then f is the zero function, which is clearly a solution. If $c = 2$, then $f(n) = 2$ when n is a perfect square and $f(n) = \pm 2$ when n is not a perfect square. The choice of the sign depends on n . To check that this is a solution, one may note that the left-hand side of (1) must be 4. The right-hand side must be $2 + 2 = 4$ for any such function. Thus, both sides are equal.

36. We label the given equation

$$f(m - n + f(n)) = f(m) + f(n). \quad (1)$$

Putting $m = n = 0$ in (1), we obtain

$$f(f(0)) = 2f(0).$$

Putting $m = 0$ and $n = f(0)$ in (1), we obtain

$$f(-f(0) + f(f(0))) = f(0) + f(f(0)).$$

Using $f(f(0)) = 2f(0)$, we have $2f(0) = 3f(0)$, i.e. $f(0) = 0$.

Putting $m = 0$ in (1), we obtain

$$f(f(n) - n) = f(n).$$

Replacing n by $f(n) - n$ in (1), we find that

$$f(m - f(n) + n + f(f(n) - n)) = f(m) + f(f(n) - n).$$

Using $f(f(n) - n) = f(n)$, this can be simplified to $f(m + n) = f(m) + f(n)$. By Cauchy equation, the possible solutions are $f(n) = cn$ for some $c \in \mathbb{Z}$. Then (1) becomes

$$c(m - n + cn) = cm + cn.$$

This holds for any $m, n \in \mathbb{Z}$ if and only if $c^2 = 2c$, i.e. $c = 0$ or $c = 2$. Thus, the solutions are $f(n) = 0$ for all $n \in \mathbb{Z}$ and $f(n) = 2n$ for all $n \in \mathbb{Z}$.

37. We label the given equation

$$f(n) + f(3m - n) = 2f(m). \quad (1)$$

Putting $m = n$ in (1), we obtain

$$f(2m) = f(m). \quad (2)$$

Putting $n = m + 1$ in (1), we obtain

$$f(m + 1) + f(2m - 1) = 2f(m). \quad (3)$$

Let $f(1) = c$. We prove that $f(n) = c$ for any $n \in \mathbb{N}$. Firstly, putting $m = 1$ in (2), we get $f(2) = c$. Putting $m = 2$ in (3), we get $2f(3) = 2f(2)$ and hence $f(3) = f(2) = c$.

Assume $f(t) = c$ for each $t = 1, 2, \dots, k$ where $k \geq 3$. If $k + 1$ is even, then we put $m = \frac{k+1}{2}$ in (2). This gives $f(k+1) = f\left(\frac{k+1}{2}\right) = c$ since $1 \leq \frac{k+1}{2} \leq k$. If $k + 1$ is odd, then we put $m = \frac{k+2}{2}$ in (3). This gives

$$f\left(\frac{k+4}{2}\right) + f(k+1) = 2f\left(\frac{k+2}{2}\right).$$

Since $1 \leq \frac{k+4}{2} \leq k$ (as $k+1$ is odd) and $1 \leq \frac{k+2}{2} \leq k$, this yields $c + f(k+1) = 2c$, i.e. $f(k+1) = c$. Thus, by induction, $f(n) = c$ for any $n \in \mathbb{N}$.

Let $n \leq 0$ and $m \geq 1$. Then (1) becomes $f(n) + c = 2c$, i.e. $f(n) = c$. Thus, f is a constant function.

Checking:

$$f(n) + f(3m - n) = c + c = 2c = 2f(m).$$

38. We label the given equation

$$f(m + n) = f(m)f(n)f(mn). \quad (1)$$

If $f(n) = 0$ for some n , then (1) becomes $f(m+n) = 0$ for fixed n and any $m \in \mathbb{Z}$. Note that $m+n$ runs through all integers. Thus, f is the constant zero function, which is clearly a solution.

Suppose $f(n) \neq 0$ for all n . Putting $m = n = 2$ in (1), we have $f(4) = f(2)^2 f(4)$. As $f(4) \neq 0$, this yields $f(2) = \pm 1$.

If $f(2) = 1$, we put $m = n = 1$ in (1). This gives $1 = f(1)^3$ so that $f(1) = 1$. Putting $n = 1$ in (1), we obtain

$$f(m+1) = f(m)^2.$$

By induction, this easily implies $f(m) = 1$ for any $m \in \mathbb{N}$. Also, this suggests f is nonnegative. Thus, using backward induction, we can prove that $f(m) = 1$ for any negative integer m . Thus, $f(m) = 1$ for any $m \in \mathbb{Z}$. This is clearly a solution.

Similarly, if $f(2) = -1$, then we follow exactly the same argument to deduce $f(m) = -1$ for any $m \in \mathbb{Z}$. One checks that this is another solution.

39. We label the given equation

$$f(m+n) + f(mn-1) = f(m)f(n) + 2. \quad (1)$$

Putting $n = 0$ in (1), we obtain $f(m) + f(-1) = f(m)f(0) + 2$.

If $f(0) \neq 1$, then we have $f(m) = \frac{2-f(-1)}{1-f(0)}$ for any $m \in \mathbb{Z}$. This shows f is a constant function. Let $f(n) = c$. Then (1) holds if and only if $2c = c^2 + 2$. There is no integer solution for c .

Therefore, we have $f(0) = 1$. In that case, we have $f(-1) = 2$. Putting $n = -1$ in (1), we obtain

$$f(m-1) + f(-m-1) = 2f(m) + 2. \quad (2)$$

Replacing m by $-m$ in (2), we get $f(-m-1) + f(m-1) = 2f(-m) + 2$. Comparing this with (2), we have $f(m) = f(-m)$ for any $m \in \mathbb{Z}$. Therefore, (2) can be rewritten as

$$f(m-1) + f(m+1) = 2f(m) + 2. \quad (3)$$

We claim that $f(n) = n^2 + 1$ for integer $n \geq -1$. The base cases $f(-1) = 2$ and $f(0) = 1$ are done. Assume this holds for $n = k$ and $n = k+1$. Using (3), we have

$$\begin{aligned} f(k+2) &= 2f(k+1) - f(k) + 2 = 2((k+1)^2 + 1) - (k^2 + 1) + 2 \\ &= k^2 + 4k + 5 = (k+2)^2 + 1. \end{aligned}$$

Thus, the case $n = k+2$ also holds. By induction, $f(n) = n^2 + 1$ for any integer $n \geq -1$. Together with $f(m) = f(-m)$, this formula holds for any integer n .

Checking:

$$\begin{aligned} f(m+n) + f(mn-1) &= (m+n)^2 + 1 + (mn-1)^2 + 1 = m^2n^2 + m^2 + n^2 + 3 \\ &= (m^2+1)(n^2+1) + 2 = f(m)f(n) + 2. \end{aligned}$$

40. We label the given equation

$$f(m+n) + f(mn) = f(m)f(n) + 1. \quad (1)$$

Putting $n = 1$ in (1), we obtain $f(m+1) + f(m) = f(m)f(1) + 1$. If $f(1) = 1$, then $f(m+1) = 1$ for any $m \in \mathbb{Z}$. This means f is the constant function 1. Clearly, it is a solution.

Suppose $f(1) = c \neq 1$. Then we have

$$f(m+1) = (c-1)f(m) + 1. \quad (2)$$

Using induction or some recursive arguments, we easily obtain

$$f(m) = c(c-1)^{m-1} + (c-1)^{m-2} + (c-1)^{m-3} + \cdots + (c-1) + 1$$

for any $m \in \mathbb{N}$. We put $m = n = 2$ in (1). Then $2f(4) = f(2)^2 + 1$ so that

$$2(c(c-1)^3 + (c-1)^2 + (c-1) + 1) = (c(c-1) + 1)^2 + 1.$$

Now,

$$\begin{aligned} & 2(c(c-1)^3 + (c-1)^2 + (c-1) + 1) = (c(c-1) + 1)^2 + 1 \\ \Rightarrow & 2c(c-1)^3 + 2(c-1)^2 + 2c = c^2(c-1)^2 + 2c(c-1) + 2 \\ \Rightarrow & 2c(c-1)^3 + 2(c-1)^2 = c^2(c-1)^2 + 2(c-1)^2 \\ \Rightarrow & (c-1)^2(2c(c-1) + 2 - c^2 - 2) = 0 \\ \Rightarrow & c(c-1)^2(c-2) = 0. \end{aligned}$$

This implies $c = 0, 2$ as we have assumed $c \neq 1$.

For $c = 0$, (2) becomes $f(m+1) = -f(m) + 1$. By induction with the base case $f(1) = 0$, it is easy to prove that $f(n) = 0$ if n is odd and $f(n) = 1$ if n is even. This is indeed in solution. If m and n have the same parity, then we have $f(m+n) = 1$ and $f(mn) = f(m) = f(m)^2 = f(m)f(n)$. If they have different parities, then we have $f(m+n) + f(mn) = 0 + 1$ and $f(m)f(n) + 1 = 1$. In any case, (1) is satisfied.

For $c = 2$, (2) becomes $f(m+1) = f(m) + 1$. By induction with the base case $f(1) = 2$, it is easy to prove that $f(n) = n + 1$ for $n \in \mathbb{Z}$. Checking:

$$f(m+n) + f(mn) = m + n + 1 + mn + 1 = (m+1)(n+1) + 1 = f(m)f(n) + 1.$$

41. We label the given equation

$$f(f(m) + f(n)) = m + n. \quad (1)$$

Note that f is injective. Indeed, if $f(\alpha) = f(\beta)$ for some $\alpha, \beta \in \mathbb{N}$, then

$$\alpha + n = f(f(\alpha) + f(n)) = f(f(\beta) + f(n)) = \beta + n$$

and hence $\alpha = \beta$.

For $n \geq 2$, we have

$$f(f(m) + f(n)) = m + n = (m + 1) + (n - 1) = f(f(m + 1) + f(n - 1)).$$

Using the injectivity, this implies $f(m) + f(n) = f(m + 1) + f(n - 1)$. Putting $n = 2$, we obtain

$$f(m + 1) = f(m) + f(2) - f(1) \quad (2)$$

for any $m \in \mathbb{N}$. Let $f(1) = a$ and $f(2) = b$. By induction, (2) easily implies

$$f(n) = (n - 1)b - (n - 2)a$$

for any integer $n \in \mathbb{N}$. Thus, (1) holds if and only if

$$\begin{aligned} & ((m - 1)b - (m - 2)a + (n - 1)b - (n - 2)a - 1)b \\ & - ((m - 1)b - (m - 2)a + (n - 1)b - (n - 2)a - 2)a = m + n \end{aligned}$$

for any $m, n \in \mathbb{N}$. Since it is a polynomial of m, n with infinitely many roots, all coefficients must be zero. We collect the like terms and find that

$$\begin{aligned} b^2 - 2ab + a^2 &= 1, \\ -2b^2 + 6ab - 4a^2 + 2a - b &= 0. \end{aligned}$$

From $b^2 - 2ab + a^2 = 1$, we have $b - a = \pm 1$. If $b = a - 1$, then $f(n) = -n + a + 1$, which is negative for large n . This is impossible. If $b = a + 1$, then the second equation can be simplified to $3a - 3 = 0$, i.e. $a = 1$. This shows $f(n) = 2(n - 1) - (n - 2) = n$ for any $n \in \mathbb{N}$.

Checking:

$$f(f(m) + f(n)) = f(m + n) = m + n.$$

42. We label the given equation

$$f(xy) = f(x)f(y) - f(x + y) + 1. \quad (1)$$

Putting $y = 1$ in (1), we have $f(x) = 2f(x) - f(x + 1) + 1$, i.e.

$$f(x + 1) = f(x) + 1 \quad (2)$$

for any $x \in \mathbb{Q}$. From this relation, we find that if $f(k) = k + 1$, then $f(k + 1) = k + 2$ and $f(k - 1) = k$. Therefore, with the base case $f(1) = 2$, we can prove by induction that $f(n) = n + 1$ for any integer n .

For any rational number $\frac{m}{n}$ where $m \in \mathbb{Z}$ and $n \in \mathbb{N}$, we substitute $x = n$ and $y = \frac{m}{n}$ in (1). This gives

$$f(m) = f(n)f\left(\frac{m}{n}\right) - f\left(n + \frac{m}{n}\right) + 1.$$

Note that $f(m) = m + 1$ and $f(n) = n + 1$. Also, we can use (2) repeatedly to obtain

$$f\left(n + \frac{m}{n}\right) = f\left(n - 1 + \frac{m}{n}\right) + 1 = \cdots = f\left(\frac{m}{n}\right) + n$$

since n is a positive integer. Then the equation becomes

$$m + 1 = (n + 1)f\left(\frac{m}{n}\right) - f\left(\frac{m}{n}\right) - n + 1.$$

After simplification, this implies

$$f\left(\frac{m}{n}\right) = \frac{m}{n} + 1.$$

Thus, $f(x) = x + 1$ for any $x \in \mathbb{Q}$.

Checking:

$$f(xy) = xy + 1 = (x + 1)(y + 1) - (x + y + 1) + 1 = f(x)f(y) - f(x + y) + 1.$$

43. We label the given equation

$$f(x + f(y)) = f(x) + f(y) + 2y. \quad (1)$$

Putting $x = y - f(y)$ in (1), we obtain

$$f(y) = f(y - f(y)) + f(y) + 2y,$$

i.e. $f(y - f(y)) = -2y$. This shows f is surjective.

Putting $x = 0$ in (1), we have

$$f(f(y)) = f(0) + f(y) + 2y.$$

From this, (1) becomes

$$f(x + f(y)) = f(x) + f(f(y)) - f(0).$$

As f is surjective, we can replace $f(y)$ by y . This shows

$$f(x + y) = f(x) + f(y) - f(0).$$

Let $g(x) = f(x) - f(0)$. Then

$$g(x + y) = g(x) + g(y)$$

for any $x, y \in \mathbb{Q}$. By Cauchy equation, the general solution is $g(x) = cx$ for some constant $c \in \mathbb{Q}$. Thus, $f(x) = cx + d$ for some constants $c, d \in \mathbb{Q}$. Then (1) is the same as

$$c(x + cy + d) + d = (cx + d) + (cy + d) + 2y.$$

This holds for all $x, y \in \mathbb{Q}$ if and only if $c^2 = c + 2$ and $cd + d = 2d$. The former equation yields $c = 2, -1$. In both cases, we have $d = 0$. Hence, the solutions are $f(x) = 2x$ for all $x \in \mathbb{Q}$ and $f(x) = -x$ for all $x \in \mathbb{Q}$.

44. We label the given equation

$$f(f(x) + y) = x + f(y). \quad (1)$$

Replacing x by $x - f(y)$ in (1), we find that f is surjective. Let a be such that $f(a) = 0$. Putting $x = a$ in (1), we obtain $f(y) = a + f(y)$, i.e. $a = 0$. This means $f(0) = 0$.

Putting $y = 0$ in (1), we obtain

$$f(f(x)) = x.$$

Replacing x by $f(x)$ in (1), we obtain

$$f(f(f(x)) + y) = f(x) + f(y).$$

So $f(x+y) = f(x) + f(y)$. This is Cauchy equation. Hence, $f(x) = cx$ for some constant $c \in \mathbb{Q}$. Then (1) becomes

$$c(cx + y) = x + cy.$$

This holds for any $x, y \in \mathbb{Q}$ if and only if $c = \pm 1$. Thus, there are two solutions, namely, $f(x) = \pm x$ for any $x \in \mathbb{Q}$.

45. We label the given equation

$$f(x + f(y)) = y + f(x + 1). \quad (1)$$

Note that f is injective. Indeed, suppose $f(\alpha) = f(\beta)$ for some $\alpha, \beta \in \mathbb{Q}$. Putting $y = \alpha$ and $y = \beta$ respectively in (1), we have

$$\alpha + f(x + 1) = f(x + f(\alpha)) = f(x + f(\beta)) = \beta + f(x + 1)$$

and hence $\alpha = \beta$.

Putting $y = 0$ in (1), we obtain

$$f(x + f(0)) = f(x + 1).$$

Using the injectivity, this implies $x + f(0) = x + 1$. It follows that $f(0) = 1$. Putting $x = -1$ in (1), we have

$$f(f(y) - 1) = y + 1. \quad (2)$$

Replace y by $f(y) - 1$ in (1). We find that

$$f(x + f(f(y) - 1)) = f(y) - 1 + f(x + 1).$$

Using (2), this becomes

$$f(x + y + 1) = f(x + 1) + f(y) - 1.$$

Replace x by $x - 1$. Then we get

$$f(x + y) = f(x) + f(y) - 1$$

for any $x, y \in \mathbb{Q}$. Let $g(x) = f(x) - 1$. The above equation becomes

$$g(x + y) = g(x) + g(y).$$

This is Cauchy equation. The solutions are $g(x) = cx$ for some constant c . Thus, $f(x) = cx + 1$ for any $x \in \mathbb{Q}$.

Now, (1) is the same as

$$c(x + cy + 1) + 1 = y + c(x + 1) + 1.$$

This holds for any $x, y \in \mathbb{Q}$ if and only if $c^2 = 1$, i.e. $c = \pm 1$. So there are two solutions, namely, $f(x) = \pm x$ for any $x \in \mathbb{Q}$.

Injective and Surjective Function

46. We label the given equation

$$f(x + y + f(y)) = f(f(x)) + 2y. \quad (1)$$

Suppose $f(\alpha) = f(\beta)$ for some $\alpha, \beta \in \mathbb{R}$. Putting $x = \alpha$ and $y = \beta$ in (1), we obtain

$$f(\alpha + \beta + f(\beta)) = f(f(\alpha)) + 2\beta.$$

Swapping α and β , we get

$$f(\beta + \alpha + f(\alpha)) = f(f(\beta)) + 2\alpha.$$

Comparing these equations and using $f(\alpha) = f(\beta)$, we get $\alpha = \beta$. This implies f is injective.

Putting $y = 0$ in (1), we obtain

$$f(x + f(0)) = f(f(x)).$$

Using the injectivity, this yields $f(x) = x + f(0)$, i.e. $f(x) = x + c$ for some constant c .

Checking:

$$f(x + y + f(y)) = x + 2y + 2c = (x + 2c) + 2y = f(f(x)) + 2y.$$

47. We label the given equation

$$f(f(x + y) - x + y) = 2y. \quad (1)$$

Putting $x = y$ in (1), we obtain

$$f(f(2x)) = 2x.$$

This implies f is injective. Indeed, if $f(\alpha) = f(\beta)$ for some $\alpha, \beta \in \mathbb{R}$, then

$$\alpha = f(f(\alpha)) = f(f(\beta)) = \beta$$

and hence $\alpha = \beta$.

Putting $y = 0$ in (1), we get $f(f(x) - x) = 0$. Since f is injective, $f(x) - x$ is a constant for any $x \in \mathbb{R}$. Let $f(x) = x + c$. Then the equation becomes

$$(x + y + c) - x + y + c = 2y.$$

This holds for any $x, y \in \mathbb{R}$ if and only if $c = 0$. Thus, the only solution is the identity function.

48. We label the given equation

$$f(f(x)) + xf(x) = 1. \tag{1}$$

Putting $x = 0$ in (1), we obtain $f(f(0)) = 1$.

We prove that f is injective. Suppose $f(\alpha) = f(\beta)$ for some $\alpha, \beta \in \mathbb{R}$. Then

$$f(f(\alpha)) + \alpha f(\alpha) = 1 = f(f(\beta)) + \beta f(\beta).$$

This implies

$$\alpha f(\alpha) = \beta f(\beta). \tag{2}$$

If $f(\alpha) = f(\beta) = 0$, then we put $x = \alpha$ in (1). This gives $f(0) = 1$. It follows that $f(1) = f(f(0)) = 1$. Putting $x = 1$ in (1), we obtain

$$1 + 1 = f(f(1)) + f(1) = 1.$$

This is a contradiction. Therefore, we have $f(\alpha) \neq 0$. It follows from (2) that $\alpha = \beta$. This shows f is injective.

Putting $x = f(0)$ in (1), we obtain

$$f(1) + f(0) = f(f(f(0))) + f(0)f(f(0)) = 1. \tag{3}$$

Putting $x = 1$ in (1), we get

$$f(f(1)) + f(1) = 1. \tag{4}$$

Comparing (3) and (4), we have $f(f(1)) = f(0)$. Using the injectivity, this implies $f(1) = 0$. From (3), we get $f(0) = 1$. However, this yields $f(f(0)) = f(1) = 0$, which contradicts $f(f(0)) = 1$.

Therefore, such a function f does not exist.

49. We label the given equation

$$f(f(x) + y) = 2x + f(f(y) - x). \quad (1)$$

Replacing y by $-f(x)$ in (1), we obtain

$$f(0) = 2x + f(f(-f(x)) - x). \quad (2)$$

Replacing x by $\frac{f(0) - x}{2}$ in (2), we find that f is surjective.

Let a be such that $f(a) = 0$. For each $t \in \mathbb{R}$, let b be such that $f(b) = t + a$. Putting $x = a$ and $y = b$ in (1), we get $t + a = 2a + f(t)$, i.e. $f(t) = t - a$. This shows $f(x) = x - a$ for some constant a .

Checking:

$$f(f(x) + y) = x + y - 2a = 2x + (y - x - 2a) = 2x + f(f(y) - x).$$

50. We label the given equation

$$f(xf(x) + f(y)) = f(x)^2 + y. \quad (1)$$

Putting $x = 0$ in (1), we obtain

$$f(f(y)) = f(0)^2 + y. \quad (2)$$

Replacing y by $y - f(0)^2$ in (2), we find that f is surjective.

Consider $x, y \in \mathbb{R}$ such that $f(x) = 1$ and $f(y) = 0$. Using (1), we have $1 = 1 + y$. This means $y = 0$ and so $f(0) = 0$. Then (2) becomes $f(f(y)) = y$. Replacing x by $f(x)$ in (1), we obtain

$$f(f(x)f(f(x)) + f(y)) = f(f(x))^2 + y.$$

This yields

$$f(f(x)x + f(y)) = x^2 + y.$$

Comparing this with (1), we have $f(x)^2 = x^2$. Thus, $f(x) = \pm x$ for each x .

Suppose there exist nonzero $a, b \in \mathbb{R}$ such that $f(a) = a$ and $f(b) = -b$. Putting $x = a$ and $y = b$ in (1), we obtain $f(a^2 - b) = a^2 + b$. This yields $\pm(a^2 - b) = a^2 + b$. This contradicts $a \neq 0$ and $b \neq 0$. Thus, either $f(x) = x$ for all $x \in \mathbb{R}$ or $f(x) = -x$ for all $x \in \mathbb{R}$.

Checking: for $f(x) = \pm x$, we have

$$f(xf(x) + f(y)) = f(\pm x^2 \pm y) = x^2 + y = (\pm x)^2 + y = f(x)^2 + y.$$

51. We label the given equation

$$f(f(x) + xf(y)) = xf(y + 1). \quad (1)$$

Firstly, the constant zero function is clearly a solution. In the following, we only focus on nonzero function f . Let y be such that $f(y + 1) \neq 0$. Replacing x by $\frac{x}{f(y + 1)}$ in (1), we find that f is surjective.

Let a be such that $f(a) = 0$. Putting $y = a$ in (1), we obtain

$$f(f(x)) = xf(a + 1). \quad (2)$$

If $f(a + 1) = 0$, then (2) implies $f(f(x)) = 0$. Replacing y by $f(y)$ in (1), we get $f(f(x) + xf(f(y))) = xf(f(y) + 1)$, so that $0 = xf(f(y) + 1)$ for any $x, y \in \mathbb{R}$. As f is surjective, $f(y) + 1$ can be any real number. This yields $xf(y) = 0$ for any $x, y \in \mathbb{R}$. This contradicts the fact that f is not identically zero by putting $x = 1$.

If $f(a + 1) \neq 0$, then we can prove that f is injective using (2). Indeed, if $f(\alpha) = f(\beta)$ for some $\alpha, \beta \in \mathbb{R}$, then

$$\alpha f(a + 1) = f(f(\alpha)) = f(f(\beta)) = \beta f(a + 1)$$

and hence $\alpha = \beta$.

Now, we put $x = 1$ in (1). This gives $f(f(1) + f(y)) = f(y + 1)$. Using the injectivity, we obtain $f(1) + f(y) = y + 1$. Putting $y = 1$, we get $f(1) = 1$. Thus,

$$f(y) = y + 1 - f(1) = y.$$

Checking:

$$f(f(x) + xf(y)) = x + xy = x(y + 1) = xf(y + 1).$$

52. We label the given equation

$$f(f(x)^2 + f(y)) = xf(x) + y. \quad (1)$$

Replacing y by $y - xf(x)$ in (1), we find that f is surjective. Suppose $f(\alpha) = f(\beta)$ for some $\alpha, \beta \in \mathbb{R}$. By putting $y = \alpha$ and $y = \beta$ in (1) respectively, we have

$$xf(x) + \alpha = f(f(x)^2 + f(\alpha)) = f(f(x)^2 + f(\beta)) = xf(x) + \beta.$$

Hence, $\alpha = \beta$. This shows f is injective.

Let a be such that $f(a) = 0$. Putting $x = a$ in (1), we obtain

$$f(f(y)) = y.$$

Replacing x by $f(x)$ in (1), we get

$$f(f(f(x))^2 + f(y)) = f(x)f(f(x)) + y.$$

Thus,

$$f(x^2 + f(y)) = f(x)x + y.$$

Comparing this with (1), we find that

$$f(f(x)^2 + f(y)) = f(x^2 + f(y)).$$

Using the injectivity, we have $f(x)^2 + f(y) = x^2 + f(y)$. This implies $f(x) = \pm x$ for any x .

Suppose there exist nonzero $b, c \in \mathbb{R}$ such that $f(b) = b$ and $f(c) = -c$. Putting $x = b$ and $y = c$ in (1), we obtain $f(b^2 - c) = b^2 + c$. This yields $\pm(b^2 - c) = b^2 + c$. This contradicts $b \neq 0$ and $c \neq 0$. Thus, either $f(x) = x$ for all $x \in \mathbb{R}$ or $f(x) = -x$ for all $x \in \mathbb{R}$.

Checking: for $f(x) = \pm x$, we have

$$f(f(x)^2 + f(y)) = f(x^2 \pm y) = \pm x^2 + y = xf(x) + y.$$

53. We label the given equation

$$f(xf(z) + y) = zf(x) + y. \tag{1}$$

Suppose $f(\alpha) = f(\beta)$ for some $\alpha, \beta \in \mathbb{R}$. Then by putting $z = \alpha$ and $z = \beta$ in (1) respectively, we have

$$\alpha f(x) + y = f(xf(\alpha) + y) = f(xf(\beta) + y) = \beta f(x) + y.$$

Note that f cannot be identically zero. Therefore, by choosing x such that $f(x) \neq 0$, we get $\alpha = \beta$. Thus, f is injective.

Putting $y = 0$ and $z = 1$ in (1), we obtain $f(xf(1)) = f(x)$. Using the injectivity, this implies $xf(1) = x$ for any x . Putting $x = 1$, we get $f(1) = 1$. Putting $z = 1$ in (1), we have

$$f(x + y) = f(x) + y.$$

Swapping x and y and comparing with the same equation, we get

$$f(x) + y = f(x + y) = f(y) + x.$$

Therefore, $f(x) - x = f(y) - y$. This shows $f(x) - x = c$ for some constant c . As $f(1) = 1$, the only possibility is $c = 0$. Thus, f is the identity function.

Checking:

$$f(xf(z) + y) = xz + y = zf(x) + y.$$

54. We label the given equation

$$f(y + zf(x)) = f(y) + xf(z). \quad (1)$$

Firstly, the constant zero function is clearly a solution. In the following, we only focus on nonzero function f . Let z be such that $f(z) \neq 0$. Replacing x by $\frac{x - f(y)}{f(z)}$ in (1), we find that f is surjective.

Suppose $f(\alpha) = f(\beta)$ for some $\alpha, \beta \in \mathbb{R}$. Then by putting $x = \alpha$ and $x = \beta$ in (1) respectively, we have

$$f(y) + \alpha f(z) = f(y + zf(\alpha)) = f(y + zf(\beta)) = f(y) + \beta f(z)$$

and hence $\alpha = \beta$ since $f(z) \neq 0$ for some $z \in \mathbb{R}$. Thus, f is injective.

Put $x = 1$ in (1). This gives

$$f(y + zf(1)) = f(y) + f(z). \quad (2)$$

Swapping y and z and comparing with (2), we get

$$f(z + yf(1)) = f(y) + f(z) = f(y + zf(1)).$$

Using the injectivity, this implies $z + yf(1) = y + zf(1)$ for any $y, z \in \mathbb{R}$. Putting $y = 1$ and $z = 0$, we get $f(1) = 1$. Then (2) becomes

$$f(y + z) = f(y) + f(z). \quad (3)$$

From this, the equation (1) becomes

$$f(zf(x)) = xf(z). \quad (4)$$

We choose x such that $f(x) = 1$. Then we obtain $f(z) = xf(z)$. As $f(z) \neq 0$ for some $z \in \mathbb{R}$, we get $x = 1$. This means $f(1) = 1$. Putting $z = 1$ in (4), we obtain $f(f(x)) = x$. Replacing z by $f(z)$ in (4), we find that $f(f(z)f(x)) = xf(f(z)) = xz = f(f(xz))$. Using the injectivity, this implies

$$f(x)f(z) = f(xz) \quad (5)$$

for any $x, z \in \mathbb{R}$. The only functions which satisfy the Cauchy-like conditions (3) and (5) are the zero function and the constant function. We have shown that the former is a solution.

Checking: for $f(x) = x$, we have

$$f(y + zf(x)) = y + zx = f(y) + xf(z).$$